

ENVIRONMENTAL SUSTAINABILITY IN HUMANITARIAN LOGISTICS



Enabling humanitarian logistics partners to reduce their impact on the environment

Improving Humanitarian Warehousing for Sustainability, Efficiency, and Cost Savings

WREC Global Info Session

June 17, 2026



Agenda

- Circular and Sustainable Warehousing Measures
- Best practice from the field
- Q&A

When you think of
circular and
sustainable
warehousing, what
ideas or practices
come to mind?



Circular and Sustainable Warehouse



A circular warehouse minimizes environmental impact by using resources efficiently, reducing waste, and keeping materials in use while supporting operations.



IMPROVING

through better structural design, energy-efficient lighting, and enhanced inventory management systems;

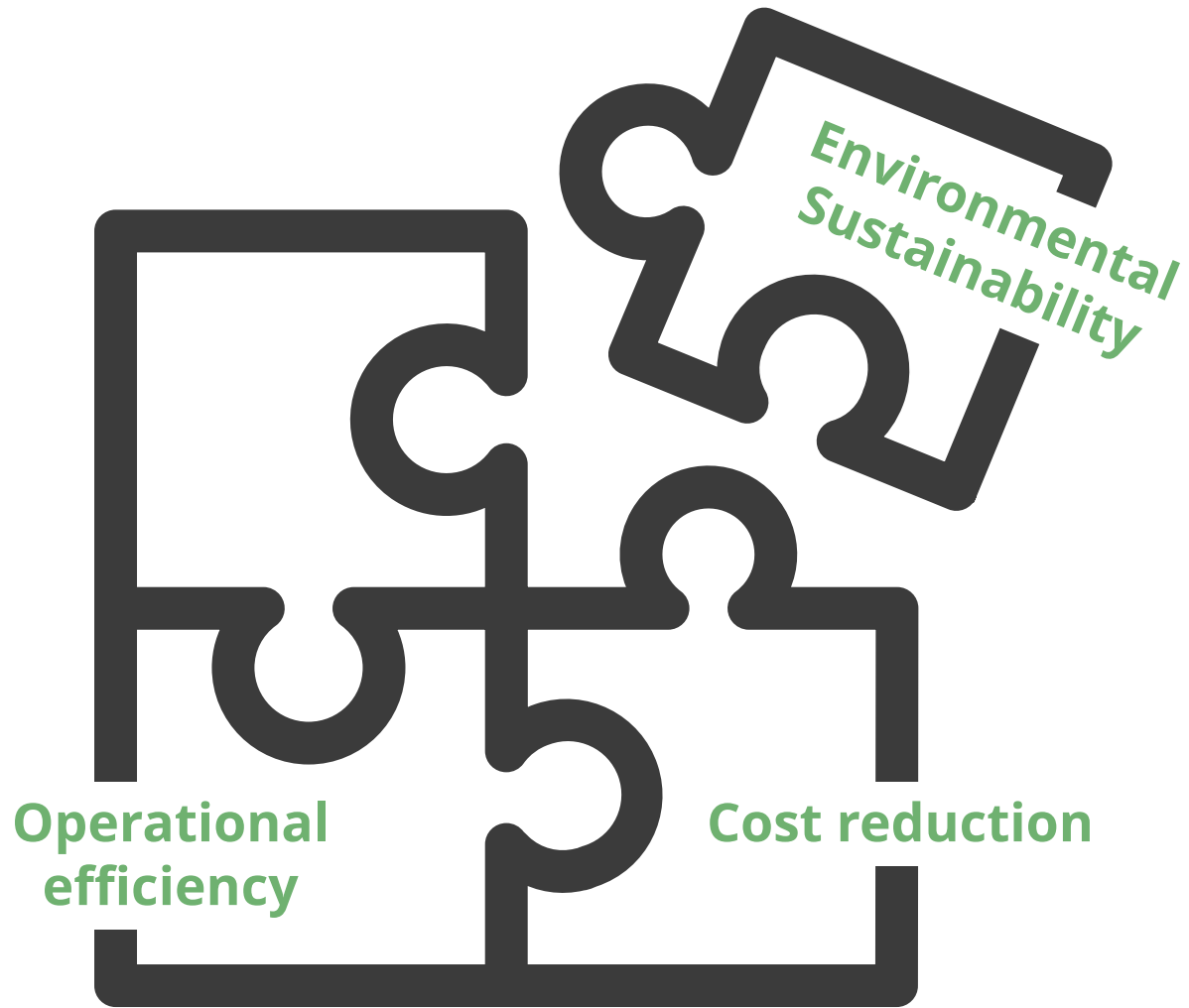
REDUCING

by minimizing water and energy consumption;

RECYCLING

by segregating and channeling waste streams such as cardboard, plastics, and metals into appropriate local recycling systems.

Circular and Sustainable Warehouse



- Circularity and environmental sustainability is not an extra layer, but a key piece of the puzzle that helps the supply chain work better
- Circularity and Environmental sustainability as a powerful operational and financial opportunity

Circular and Sustainable Warehouse



Structure



Energy



Storage & Space optimization



Waste



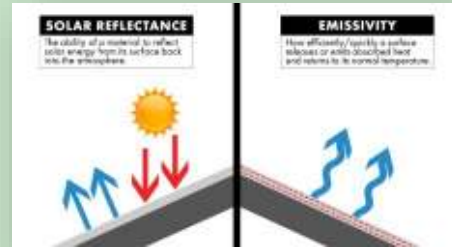
Behavioral change



Insulation of walls and roofs



Reflective or cool roofs



Solar shading



Passive ventilation and cooling



Recycled or low-carbon material



Water harvesting infrastructure and Install low flow fixtures and repair leaks promptly





Switch to efficient LED lighting



Choose energy efficient equipment



Install motion sensors and remote switches



Install renewable energy systems



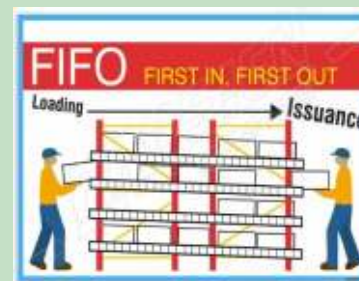
Storage and space optimization



Optimize space use



Adopt a First In, First Out (FIFO) or First Expired, First Out (FEFO) approach





Know your waste

	Weekly			
Composition	Net (KG)	%	Volume (L)	%
Plastic material	43	12%	7 bins * 120L	
Paper and cardboard material	301	81%	12 bins * 120L	
Organic material	25	7%	1 bin * 120L	
Other	1	0.3%	33% bin * 120L	
TOTAL sample after sorting	370	100%	21 bin 120L	



Explore local options for each waste stream



WREC Waste Management & Recycling Infrastructure mapping

Implement source segregation, proper storing, and safe handling of waste



WREC Waste Measurement Decision Tree

Behavioral change



Training



Awareness and reminder tools



Which of the measures
we discussed today is
most applicable to
your warehouse?



Any additional
measures that were
not covered?





WFP Chad

Ingrid Beyegue

Circular & Sustainable Warehousing Innovation

Temperature-controlled warehouse and modular shelving built with recycled MSU materials

Bagarine, Abéché | Ouaddaï, East Chad
WFP Chad Country Office

Prepared using available project documentation.
Thermal mapping is positioned as a next step.



Circular Economy logic

From damaged and unused MSU components to operational Warehouse infrastructure*



Inventory and material assessment from MSU poles and metallic tubes



Circular logic

Recover → Redesign → Build → Reuse in operations

The approach extends material life while reducing the need for new structural components.



Recovered MSU components assessed for reuse

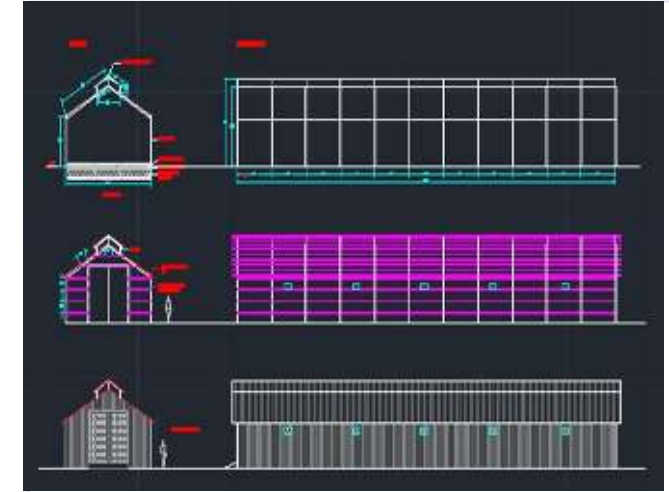
* MSU : Mobile Storage Unit



Recycled steel frame and roofing works

Passive cooling approach

- Two-segment roof with a 0.6m ventilation gap.
- Aluminum-zinc external wall cladding.
- Internal lining with cellulose foam and plywood.
- Painted internal finishes for usability and durability.



Foundation / early construction works

Warehouse configuration

- 10m wide, 44m long and 3.50m high.
- Structure built using recycled old metallic bars from MSUs.
- Five ventilation windows on each side, each 0.4m x 0.6m.
- Two main doors of 5m x 4m to allow large truck access.
- Reinforced concrete foundation with starter columns and base plates for the main vertical frames.

1

**Material Inventory
Assessment &
design**

2

**Foundation
works**

3

**Frame
erection**

4

**Roofing
& ventilation**

5

**Cladding
& interior finish**



Foundation and floor
reinforcement



Frame erection



Wall cladding / ventilation view



Interior finishing and lining



Assembly of modular shelving components

Load-testing with heavy materials
Approx. 500 Kg per compartment

Shelving features

- Built from recycled MSU metallic parts.
- Approximate dimensions: 3m high x 7.5m long.
- Detachable design with three compartments.
- Can be disassembled, transported and reassembled based on warehouse needs.
- Designed for practical assembly by non-specialized labour.



Why a thermal mapping and what it can answer ?

- How temperature varies between entrance, center, walls and roof-adjacent zones.
- Whether higher layers or shelf levels experience greater heat exposure.
- Whether design features are producing measurable improvements compared with standard structures.
- What monitoring configuration should be maintained for sensitive commodities.

Expected outputs

- Validated temperature-performance profile.
- Evidence-based communications package for circular warehousing.
- Operational recommendations for commodity placement and monitoring.
- Replication considerations for other hubs and field operations.

Location

Chad, Bagarine, Abeche

Period

July 2024 – April 2025

Warehouse size

10m x 44m x 3.5m

What the project demonstrates ?

- Circular use of MSU metallic parts that were originally intended for disposal.
- Construction of a 10m x 44m warehouse with passive cooling features designed to improve storage-condition.
- Creation of modular heavy-duty shelving using the same recycled MSU material stream.
- A practical example of field-level innovation in humanitarian logistics infrastructure.

What will come next ?

- Shelf load-capacity
- thermal mapping and temperature-performance study



Source: Chad Engineering team project brief, May 2025 | Prepared for experience sharing

Already documented

- Project location, timeline and engineering ownership.
- Material-reuse concept using MSU components originally intended for auction.
- Warehouse dimensions, structural elements and passive cooling design features.
- Construction sequence and visual evidence from the project brief.
- Modular shelving concept, dimensions and load-testing narrative.

To be documented through next steps

- Measured temperature profile inside the warehouse over time.
- Temperature stratification by zone and height inside the facility.
- Comparison with a standard MSU or conventional warehouse under similar conditions.
- Validation of any temperature reduction claim and operational benefits.
- Confirmed current commodities stored and user feedback from Abeche operations.

“A promising circular warehousing solution, with thermal performance documentation planned as the next evidence-generation step.”



World Food
Programme

SAVING
LIVES
CHANGING
LIVES

Thank you



- ✓ Abeche project demonstrates a practical circular warehousing model
- ✓ Thermal mapping will provide the additional layer of measurable evidence.

WFP Chad Country Office | Prepared with available project information



Danish Refugee Council

*Tamby Libzo & Randa
Alattar*





UNICEF

Judith Okoli

Improving Warehousing for Sustainability, Efficiency, and Cost Savings

Bauchi Warehouses Solar PV Case Study

Demonstrating how sustainability drives operational performance, resilience and cost reduction

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Bauchi Warehouses Case Studies

- 3 WH sites assessed for solar PV implementation
- Sites: Galadima, Fertilizer and Gidan Gona Warehouses
- Leveraged GIZ support for energy audits pro-bono
- Objective: Improve sustainability, reduce emissions & environmental impact, achieve cost savings and strengthen operational resilience.
- Systems sized based on actual load profiles

Site Characteristics – UNICEF Bauchi Warehouse – Gidan Gona

The building is located in Bauchi, Bauchi State

Total roof area available: 300m²

Roof area available for PV system: 300m²

Ground space available for PV system: N/A

Site Coordinates: 10.307965, 9.853729

Current energy supply: Grid only



Galadima Warehouse: High Impact Investment

-  17 kWp PV + 20 kWh battery
-  83.4% solar coverage
-  Annual PV production: 29,397 kWh
-  IRR: 17.1%
-  Payback period: 5.54 years
-  85% reduction in diesel consumption
-  78% reduction in CO₂ emissions

Site Characteristics – UNICEF Bauchi Warehouse - Galadima

The building is located in Bauchi, Bauchi State

Total roof area available: 5000m²

Roof area available for PV system: 5000m²

Ground space available for PV system: N/A

Site Coordinates: 10.305233,9.792803

Current energy supply: Grid , 2 x Diesel Genset only



Bauchi Warehouses Case Study Overview

- Galadima: 17 kWp + 20 kWh battery | 83.4% solar coverage | US\$23,650
- Fertilizer: 2.2 kWp + 7 kWh battery | 69.6% solar coverage | US\$5,754
- Gidan Gona: 2.2 kWp + 8 kWh battery | 67.0% solar coverage | US\$5,959

Site Characteristics – UNICEF Bauchi Warehouse - Fertilizer

The building is located in Bauchi, Bauchi State

Total roof area available: 500m²

Roof area available for PV system: 500m²

Ground space available for PV system: N/A

Site Coordinates: 10.395737,9.742979

Current energy supply: Grid only



Next steps, Lessons Learned & Replicable Practices



Self- Investment using global long-term arrangements with solar goods and services



Leverage partnerships; size systems based on actual load profiles



Combine sustainability objectives with operational needs



Focus on solutions that improve both resilience and cost efficiency



Use pilot projects to demonstrate value and scale up



Does your organization
have any practices in
place to make your
warehouse more
circular and
sustainable?



Q&A

THANK YOU!

FOR MORE INFORMATION:

[HTTPS://LOGCLUSTER.ORG/WREC/GREEN-LOGISTICS](https://logcluster.org/wrec/green-logistics)

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